Ph.D. Dissertation Proposal

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August 10, 2025

Pertinent Information

Proposed Title: Acoustic Detection of Unmanned Aerial Vehicles and Insects in Stored Products Leveraging Relational Databases to support Multi-modal and Multi-sensor Data

Prospective Committee Members

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Contents

1 Introduction

Acoustic sensing has been a reliable and relatively low-cost alternative to even the most advanced sensing technologies to provide a method of detecting, localizing, tracking, and classifying subjects including cracks in wind turbine blades, marine mammals and scuba divers underwater, and low-flying aircrafts [1], [2], [3].

Large amount of acoustic data recordings are required not only to study the characteristics of the acoustic signals generated by the studied subjects, develop algorithms, validate sensor performance, but also to support the increased advancements and usage of machine learning and artificial intelligence. While the research community has access to vast repositories of text, image, and video datasets that have driven significant advances in learned language modeling (LLM) and computer vision, acoustic datasets remain comparatively scarce [4], [5], [6]. This might be attributed to visual data modalities such as images and video are easily annotated and inherently interpretable due to their direct representation of observable phenomena. Acoustic data presents a more abstract signal that necessitates corroborative information including subject characteristics, environmental parameters, and sensor specifications to enable meaningful analysis and classification.

This work focuses on the demonstration of the possibilities of acoustic sensing to detect unmanned aerial vehicles and insects within stored products by using simple signal processing techniques with promising performance and avoiding the need for machine learning and artificial intelligence. On the surface, these targets seem superficially different from their size and function to their acoustic signatures. Conversely, their audible existance with unique sonic characteristics make them opportune canditates for acoustic tracking. Furthermore, they both pose a threat to national security, drones as they become more prelevant for espionage, smuggling, and as weapons or insects as they affect food safety, agriculture, the environment, or enter delicate ecologies as invasive species.

In the first section, the acoustic detection of unmanned aerial vehicles (UAVs) is discussed. The section presents an updated prototype to the previously developed Drone Acoustic Detection System (DADS). The Drone Acoustic Recording Array (DARA) is a seven-microphone system compared to the four-microphone system of the DADS. Both sensors were deployed over the course of two field tests where numerous drones of different sizes and amount of propellors were flown. After comparing the spectrograms of the UAVs and finding their sound pressure levels (SPL) at 1 m, an algorithm was developed to predict the detection range of each sensor at different noise levels and environtal conditions using wideband processing and the sonar equation. The actual detection algorithm utilizes the Steered-Response Phase Transform (SRP-PHAT) to localize UAVs [7].

The following section examines the acoustic detection of insects in stored products. The Acoustic Stored Product Insect Detection System (A-SPIDS) was developed to address the need for an efficient alternative to manual visual inspection of materials such as grain, rice, and legumes for insects at United States Ports of Entry. The sensor was used to collect an extensive amount of recordings using live insect specimen at various life stages in materials of different kernel size and hardness with varying levels of natural and simulated background noise. This data was used to study the characteristics of the impulses generated by the insects as they manuevered around the material and derive metrics based on the pulse amplitude and spectra. A method of normalization was also introduced that could extend to other insect sensors. Utilizing these normalization and filtering techniques, a detection algorithm was developed to effectively detect the presence of a single insect within 1 L of material with up to 90 dBA external noise [8].

Underlying both of these studies is a database package that leverages relational database structures to organize and store the large amounts of acoustic data along with their respective corroborative information. The Sound Organization and Network Integration for Collection/Collaboration Database (SONICDB) was designed to be flexible and extendable, allowing for its adoption to service projects of varying needs. The Stored Product Insect Database (SPIDB) is a simple extension of

the SONICDB package, only adding models to support logging of the insect subjects in their paired materials. In the Drone Acoustic Dataset, the SONICDB package supports data from five field tests, incorporating acoustic data from 14 different sensors recording a total of eight different drone models. Additionally, the dataset also features flight logs, GPS truth data, ADS-B, and weather data.

2 Acoustic Detection of Unmanned Aerial Vehicles

2.1 Literature Review

Unmanned Aerial Vehicles (UAVs), also known as drones, have become prelevant in consumer, commercial, industrial, and defense industries. Although their usage has entertained families, assisted farmers to monitor and service large amounts of agricultural areas, and enabled first responders, drones have also been used in nefarious purposes such as surveillance, smuggling, and as weapons [9]. The weaponization of UAVs has been highlighted by the conflict in the Russian-Ukraine war demonstrating the efficacy of low-cost drones to monitor the battlefield, deliver explosives, jam communications, and even conduct kamikaze-style attacks [10]. To combat this threat, Counter-Unmanned Aerial Systems (C-UAS) have been developed to detect, localize, track, classify, and mitigate using sensing technologies including radar, passive radio-frequency, electro-optical (EO) and infrared (IR) cameras, and acoustics. Although acoustic sensors provide a relatively lower detection distance than their radar and RF counterparts, they are comparitevely lower cost, passive, and are capable of detecting, localizing, tracking, and classifying UAVs [11].

Diaconu and Petrița demonstrated that a standard off the shelf sound card with a four microphones array successfully estimated the direction of arrival within 5° [12].

2.2 Motivation

As drone usage becomes more pervasive, the need for effective detection and tracking systems has grown. Although the relatively low detection distance and susceptibility to environmental noise are a challenge for acoustic sensors, they still offer a promising solution due to their low cost, passive nature, and weather resistance especially when deployed in a networked and fusioned system with other modalities. By publishing this research as well as the data collected, it is hoped that it will encourage further development and refinement of algorithms, platforms, and usage within machine learning models.

2.3 Methodology

3 Acoustic Detection of Insects in Stored Products

3.1 Literature Review

Stored products refer to dried, durable raw agricultural and processed commodities such as grains, cereals, seeds, nuts, and dried fruits that are kept for extended periods and make up the largest percentage of plant based food [13]. These products are susceptible to infestation by stored product insects during farming, storage, and transportation.

Insects attack stored products in a variety of ways. The rice weevil, *Sitophilus oryzae*, the granary weevil, *Sitophilus granarius*, and the lesser grain borer, *Rhyzopertha dominica*, are an example of internal feeders, which bore into the kernels of grains and seeds as larvae and feed on the endosperm, visually undetectable, until they emerge as adults leaving burrowing holes and powder residue. External feeders, such as the drugstore beetle, *Stegobium paniceum*, the cigarette beetle, *Lasioderma serricorne*, and the Khapra beetle, *Trogoderma granarium*, chew through the outer seed coat of the grain exterior and then devour the inside. The red flour beetle, *Tribolium castaneum*, and the sawtoothed grain beetle, *Oryzaephilus surinamensis* are examples of scavenger feeders, which feed on broken down and processed material such as broken kernels and flour. Secondary pests, like the mealworm, *Tenebrio molitor*, eat deteriorating, damp, or moldy material [14].

An unmitigated infestation can lead to significant ecological, agricultural, and economic impact. Post-harvest insects are responsible for 10 % to 20 % of storage losses. Beyond the direct loss of material, which can range from 10 % to 25 % depending on the pest species and storage conditions, the insects can introduce pathogens and allergens into the material, degrade the quality of the product by changing their chemical and nutritional composition, and produce metabolic products that further degrade the material. Invasive species, such as the Khapra beetle, can enter new regions and disrupt the local ecosystem and economy. The presence of insects in the material can also lead to the rejection of the product by the consumer, the importer, or the regulatory agency [13], [14], [15].

3.1.1 Methods for Stored Product Insect Detection

A container is deemed infested if a single insect is found within 1 kg of material [16]. The main method employed at United States Ports of Entry to inspect stored product for insects is through manual inspection, where the chosen commodity is emptied on an inspection table and personnel, often with tools including sieves, tweezers, magnifying glasses, and intense lights sift through the material looking for live and dead insects at different life stages, fragments, or signs of insect infestation such as webbing, casings, frass, or burrowing holes in the kernels. This process is time and physically consuming, requiring experienced personnel or with a background in entomology to identify the species of the insect in order to determine the appropriate course of action. If the insect is identified as a quarantine pest, the material is either treated, destroyed, or rejected and a Emergency Alert Notification (EAN) is issued to the importer [17].

Various tools and devices have been developed, often leveraging sensing technologies, in order to expedite areas within the inspection including automating the examination process, detecting insect infestation, or identifying the insect species. Physical methods use probe traps and counters, such as the Berlese-Tullgren trap, the TNAU Insect Detection System, and the Electronic Grain Probe Insect Counter (EGPIC), to capture and count the manuevering material to determine the level of infestation using extrapolation. Although visual lures and pheromones assist in attracting insects to the traps, these physical methods require a long time duration and cannot detect insects buried inside kernels [18], [19]. Other physical methods include rolling machines that automate the sieving

of products, systems that monitor for changes in electrical conductance between whole, damaged, and infested kernels, and the flotation method which separates possible insects from product by density, but these tend to damage the material in the process [20], [21], [22].

Chemical sensors monitor the O_2 and CO_2 levels in the stored product to detect the respiration of the insects or by measuring for uric acid, the main element of insect exreta [23], [24], [25]. Spectral imaging use near-infrared (NIR), thermal, hyperspectral, and X-ray imaging to detect and classify the infestation [26], [27], [28]. Although machine learning image recognition algorithms have effectively been used to classify the insect species, these systems are still expensive and require highly trained personnel to operate.

Acoustic sensors leverage traditional microphones, piezoelectric discs, and accelerometers to detect the sound waves and vibrations generated by the insects as they move, communicate, masticate, and procreate in the material [8], [29], [30], [31]. These systems are designed as standalone receptacles where product is placed momentarily during examination or as probes and sensors that could be inserted into the product or mounted on the storage containers even for continuous monitoring. Microwave systems track the movement of insects through a material by detecting differences between emitted and reflected radar frequencies [32]. Although both these methods are relatively low-cost and non-destructive, they are highly succeptible to external environmental noise and vibration.

3.1.2 Acoustic Algorithms for Stored Product Insect Detection

Acoustic algorithms for stored product insect detection can be classified into two overarching categories. Historically, signal processing and statistical modeling based algorithms utilized the characteristics of the audio signals generated by the insects to determine the presence and severity of the infestation. More recently, machine learning algorithms have been employed to leverage the emerging technology of artificial intelligence and deep learning capabilities towards the detection process. These algorithms can be tasked to detect the presence of insects, classify the species and lifestage, determine the level of infestation, and identify the behavior of the subject [33]. Often the signal features, processing, and algorithms are common with the analysis and detection methods for insect infestations in other materials such as plants, soil, and wood [34].

Insect activity manifests in acoustic recordings as impulses, high magnitude signals across a wide frequency range and short duration. Insects generate sounds within the material during locomotion through friction between the body and the material as well as through the movement of the wings, legs, and other body parts. Sounds are also generated by mastication, communication, and procreation. The production of these sounds is strongly affected by the insect size and lifestage, the density of the infestation, and behavior of the subject which can vary with temperature and external disturbances. The propogation of the sound is dependent on the material properties including density, kernel size and hardness, relative humidiy, and the amount of air in the material [35]. The distance between the insect and the sensor as well as variation in the recording equipment and environmental conditions can introduce differences in the results from different studies and publications. Normalization methods can be employed to reduce the effect of these variations and to allow for the comparison of results between different studies [8].

Detection algorithms analyze and use the spectral and temporal characteristics of the signals to determine the presence of insects. The spectral domain refers to the frequency content of sound. Generally, insect activity occurs in the frequency band between 80 Hz to 8000 Hz [35] and can vary between the species and lifestages [36], [37]. Schwab et al. were able to distinguish between the larval and adult stages of the *Sitophilus granarius*, *Tribolium confusum*, and the *Rhyzopertha dominica* through spectral features alone. The temporal domain refers to the time-varying nature of the signal including amplitude and duration. Although a single impulse might be 3 ms to 30 ms,

insect bursts are characterized as a series of patterns with a duration of 250 ms. Some algorithms count the number of bursts and impulses to determine an infestation [29], [30], [35], [39].

Signal processing such as filters and transformations can be used to isolate the frequency range of interest and expose the relevant signals while removing superfluous noise from analysis that can vary between sensors and environments. The waveform envelope allows for threshold crossing algorithms to determine an insect precense if the amplitude exceeds a statitically or empirically determined cutoff [30]. Mankin et al. observed that the sounds of *Sitophilus oryzae* movement and feeding were 23 dB. Balingbing et al. continously recorded the *Rhyzopertha dominica*, the *Sitophilus oryzae*, and the *Tribolium castaneum* in rice over a 24 h time frame using an Adafruit 12S MEMS SPH0645 microphone and calculated the root-mean-square (RMS) of 3 min windows at each hour of a signal filtered between 93.75 Hz to 2500 Hz. Eliopoulos et al. utilized a 14 cm long piezoelectric sensor inserted with a probe into hard wheat grain. The system was able accurately detected the presence of an insect with a probability range from 72 % to 100 %, a 100 % false positive rate. Their algorithm was based on extracting the Hilbert envelope and indicating a detection when it meant a certain threshold.

The regularity and duration of the monitoring also has an effect on the performance of the sensor and its algorithm. Vick et al. noted that the *Rhyzopertha dominica*, *Sitotroga cerealella*, and *Sitophilus oryzae* produced sounds 61 %, 71 %, and 90 % of the time in grain during sampling durations of 180 s. Short analysis durations can increase high rates of false negatives especially if the insects are not active during the sampling period. Longer sampling periods, such as 10 min to 20 min allow for a more accurate detection as well as enable identification of background noise [35]. Deployment of multiple sensors simultaenously could increase detection performance, determine locations of insects, level of infestation, and to offset the frequency and duration of monitoring [35]. Hagstrum et al. concluded that the frequency of monitoring a single sensor had an equivalent performance as monitoring multiple sensors at a lower frequency.

Limiting the influence of external noise and vibration is critical to the success of these algorithms. Often background noise can be reduced physically through shielding, sound-proofing, and vibration isolation. This could be through box in a box design, anechoic chambers, employing Faraday cages, and shock mounts. Alternatively, reference sensors could be used to measure and identify peripheral noise and vibration [29]. The spectral and temporal differences between the insects and background noise can also be used to distinguish between the two. Mankin et al. used filtering to reduce signals below 200 Hz. In another algorithm, the bursts characteristic to distinguish the insect sounds from the continuous or isolated impulses commonly found in background noise [45], [46].

Machine learning algorithms can analyze acoustic signals and identify patterns associated with insect activity as well as distinguish external noise however they depend on large amounts of training data. The ML models can be trained on the raw audio signals or on extracted features and the data can be provided labelled or unlabelled [33]. These algorithms include various nueral network classifiers such as convolution neural networks (CNN), probabilistic neural networks (PNN), perceptual learning prediction (PLP), decision trees and forests, hidden Markov models (HMM), support vector machines (SVM), and bayesian classifiers [30]. Deep learning combines multiple layers of neural networks to learn the features and patterns of the data. Since machine learning algorithms are data-driven, the performance of the model might not generalize to new species or materials, if the training data is not diverse enough.

Phung et al. used the BugBytes dataset to compare training a linear discriminant, fine kNN, linear SVN, and Bagged Tree model using low-level features such as the zero-crossing rate, short-time average energy, bandwidth, and cut-off frequency as well as the mel-frequency ceptstrum coefficients (MFCC). They concluded that the Bagged Tree model provided the best accuracy for species classification and insect classification with an accuracy of 97.1 % and 92.3 %, respectively while using both low-level and MFCC features. Montemayor et al. used the mel-frequency cepstrum coefficients from

the audio signals and their log-scaled representations as Time-Frequency Patches (TF-Patches) as inputs to a Convolutional Neural Network (CNN). Faiß and Stowell observed that training a LEAF model with raw waveform data from the diverse InsectSet32, InsectSet47, and InsectSet66 resulted in better performance than a CNN model with spectrograms, a graphical visualization of the frequency magnitude over time.

3.1.3 Acoustic Insect Datasets

Several audio databases featuring insects have been published and are publically available. The Animal Sound Archive [50], InsectSet32 [51], InsectSet47 and InsectSet66 [49], [52], InsectSound1000 [53], and Singing Insects of North America (SINA) [54] all feature insect recordings; however, these datasets feature neither species commonly found infesting stored products nor audio recordings of insects within materials or stored product. BugBytes features 95 recordings of insects in different environments but only presents approximately 23 minutes of recordings with less than 15% of that pertaining to stored product insects; furthermore, only two of those recordings use piezoelectric disks similar to the systems we use for PestPulse for additional sensitivity and measurement [55].

3.2 Motivation

The motivation behind this research is to enhance the detection and monitoring of insects in stored products mainly to assist and speed up inspections performed by Custom and Border Patrol (CBP) at United States Ports of Entry. Although the sensors and associated algorithms were designed for this usage, the platforms can be adapted monitoring for stored product insects at other locations in the supply chain. The underlying technology and methods can also be expanded to service other materials. Beyond ensuring food safety and quality, ensuring effective monitoring can secure national agricultural, environmental, and ecological security from invasive species.

3.3 Methodology

3.3.1 Acoustic Stored Product Insect Detection System (A-SPIDS)

The Acoustic Stored Product Insect Detection System (A-SPIDS) was designed as a low-cost, non-destructive, and portable alternative inspection system. The platform utilizes a box-in-a-box design where a sample test container capable of handling approximately 1.5 L of dry stored product is housed within a larger chamber lined with silicon and acoustic foam to physically reduce external noise and vibration. The bottom of the testing recepticle is lined with 32 piezoelectric discs. The 27 mm discs are arranged in a four groups of eight and convert the mechanical vibrations generated by the insects into electrical signals. The four groups correspond to channels zero through three in the recordings.

Traditional acoustic microphones, which transmit through channels four to seven, are placed throughout the body of the system to capture the external noise and vibration. Microphone channels four and five are colocated with the piezoelectric channels zero and one, respectively, while channel six is placed within the acoustic foam, and channel seven is mounted on the exterior of the system. Electronics, including the data aquisition system and batteries, are located in the base of the platform. Data is transmitted wirelessly to a computer for recording and analysis. Further details on the system design are described in a previous publication [8].

3.3.2 Tested Insect, Materials, and Noise Levels

Although stored products insects proliferate across a wide range of commodities, they are difficult to obtain for research purposes. The cowpea beetle, *Callosobruchus maculatus*, the confused flour beetle, *Tribolium confusum*, and the mealworm beetle, *Tenebrio molitor*, are commercially available and were chosen based on their similarities to the sizes, qualities, and behavior to the species targeted during inspection.

A variety of stored product materials were tested with and without insect specimen including oatmeal, *Avena sativa*, rice, *Oryza sativa*, wheat in the form of whole groat kernels and processed flour, *Triticum aestivum*, and corn in the form of Corn Flakes, *Zea mays*. The material samples were kept in sealed storage containers to prevent contamination and infestation. During testing, 1 L of clean material was transferred into a plastic bag and placed into the sensor container. After several minutes of silence were recorded to ensure the grains were settled, a single insect was introduced into the material.

Testing was conducted in a varying levels of ambient and simulated noise. The ambient noise was a result of the surrounding laboratory environment including equipment, HVAC units, as well as vehicular and aircraft traffic. Artificial noise, such as pre-recorded sweeps, white noise, loud conversations, crowds, and factory noise was played through a speaker system and the noise levels were measured using a calibrated sound level meter. Further information on the tested insects, materials, and noise levels can be found in the previous publication [8].

3.3.3 Stored Product Insect Database (SPIDB)

The Stored Product Insect Database (SPIDB) provides interface to access the data collected from the A-SPIDS platform. The database extends the Sound Organization for Network Integration and Collection/Collaboration (SONICDB) framework, a Python package developed to store, organize, and retrieve multi-modal acoustic data and associated metadata. The SONICDB package uses popular and standard libaries and is designed to be extensible and customizable. The Stored Product Insect Database adds the `Material` model and additional attributes to the `Event` and `Subject` models. Figure **??** shows the schema for the SPIDB:

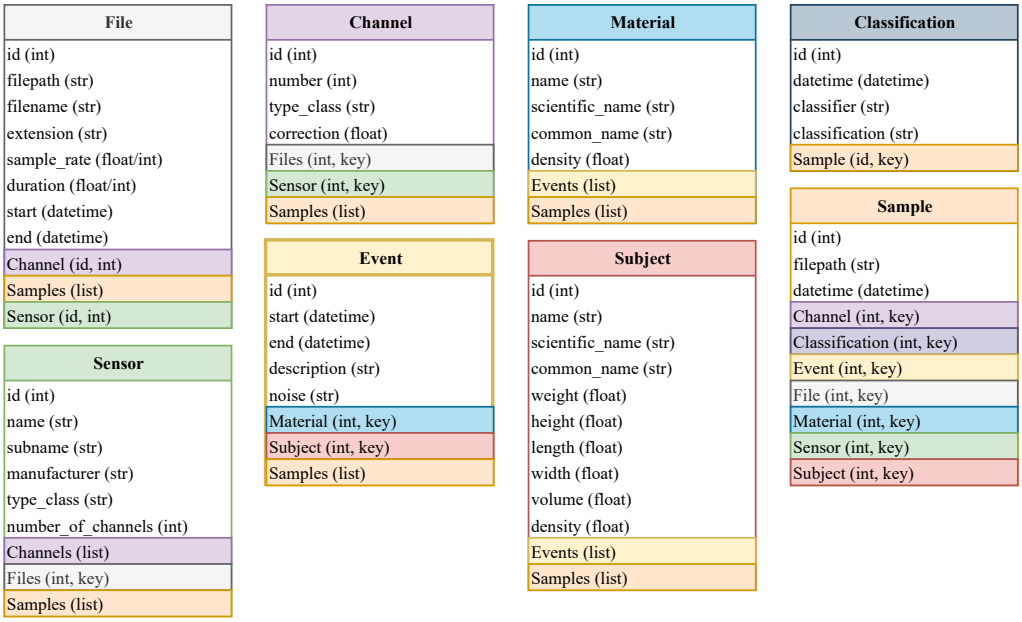

Figure 1: Schema of the Stored Product Insect Database (SPIDB)

The Stored Product Insect Database (SPIDB) is available on GitHub with supporting documentation [56]. The associated dataset is available on Kaggle [57]. The dataset can be used to further develop and test acoustic detection algorithms, train machine learning models, and explore the acoustic signatures of other insects and materials. Since obtaining and maintaining live insect specimens for recording requires financial and time commitment, efforts can be made to extend the database. The recordings can be augmented using techniques such as noise injection, volume control, fading, pitch shifting, time shifting, time stretching, temporal inversion, clipping, and distortion [58]. Algorithms can be developed to generate synthetic data simulating different insect sizes and life stages, materials, and noise levels. The recording and synthetic data can be used to further train machine learning models on different insect species and life stages, materials, and noise levels.

4 Back Matter

4.1 Research Timeline

Completed Tasks

- **February-May 2022** Literature review and analysis of acoustic detection data from 2021 and 2022 field tests. Preparation for conference presentation and publication.
- **March 2022** Ph.D. Qualification Exam
- **May 2022** ASA Conference, Denver, CO, USA - Improvements to the Stevens Drone Acoustic Detection System
- **February-December 2023** Development of the Acoustic Stored Product Insect Detection System (A-SPIDS) and collection of vibro-acoustic signatures of various insects in stored products.
- **January 2024** Literature review and analysis of vibro-acoustic signatures of various insects in stored products. Preparation for publication.
- **March 2024** AIMS Conference, New York, NY, USA - Perspectives of Machine Learning for Acoustic Methods of Stored Product Insect Detection
- **January 2025** Publication of *Vibro-Acoustic Signatures of Various Insects in Stored Products* in MDPI Sensors
- **January-August 2025** Development and validation of stored product insect detection algorithm
- **August 2025** DHS RDT&E Conference, Washington, D.C., USA - Stored Product Insect Detection System (SPIDS)

Current Tasks

- **August 2025** Dissertation Proposal Defense
- **September 2025** Submit *Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions* to MDPI Sensors
- **October 2025** Submit *The Drone Acoustic Database: A Rich Multi-Sensor Dataset for Acoustic UAV Detection and Classification* to MDPI Data
- **November 2025** Work on dissertation

Planned Tasks

- **September 2025** Submit *Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions* to MDPI Sensors
- **October 2025** Submit *The Drone Acoustic Database: A Rich Multi-Sensor Dataset for Acoustic UAV Detection and Classification* to MDPI Data
- **November 2025** Work on dissertation
- **End of December 2025** Dissertation Defense

4.2 Publications and Conferences

Publications

- **Vibro-Acoustic Signatures of Various Insects in Stored Products**

  D. Kadyrov, A. Sutin, N. Sedunov, A. Sedunov, and H. Salloum, "Vibro-Acoustic Signatures of Various Insects in Stored Products," Sensors, vol. 24, no. 20, Art. no. 20, Jan. 2024, doi: 10.3390/s24206736.

Planned Publications

- Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions
- The Drone Acoustic Database: A Rich Multi-Sensor Dataset for Acoustic UAV Detection and Classification

Conferences

- **Improvements to the Stevens Drone Acoustic Detection System**

  Acoustic Society of America (ASA) 2022, Denver, CO, USA

  D. Kadyrov, A. Sedunov, N. Sedunov, A. Sutin, H. Salloum, and S. Tsyuryupa, "Improvements to the Stevens Drone Acoustic Detection System," Proceedings of Meetings on Acoustics, vol. 46, no. 1, p. 045001, Aug. 2022, doi: 10.1121/2.0001602.

- **Perspectives of Machine Learning for Acoustic Methods of Stored Product Insect Detection**

  Artificial Intelligence Meets Science (AIMS) Conference, 2024, New York, NY, USA

- **Stored Product Insect Detection System (SPIDS)**

  Department of Homeland Security (DHS) Research and Development, Testing and Evaluation (RDT&E) Conference, 2025, Washington, D.C., USA

4.3 Policy and Regulatory Considerations

The research and products developed as part of the unmanned aerial vehicle study were funded by the Air Force Research Laboratory (AFRL) under contract AFRL-2022-3230. Prior to public release, all publications and work products need to be reviewed and approved by AFRL Public Affairs Office. The drone flights were conducted in accordance with the Federal Aviation Administration (FAA) regulations under Part 107.

The research and products developed as part of the stored product insect detection study were funded by the Cross-Border Threat Screening and Supply Chain Defense (CBTS) under grant 18STCBT00001. This office no longer exists and the publications and work no longer need to be reviewed and approved by this office. The insects used in the study are non-invasive and non-harmful to the environment, and the research was conducted in accordance with the Institutional Animal Care and Use Committee (IACUC) guidelines. The insects were disposed of after the completion of the study following the IACUC and USDA guidelines.

Daniel Kadyrov is currently a full-time employee of the National Urban Security Technology Laboratory (NUSTL), within the Science and Technology Directorate (S&T), a part of the Department of Homeland Security (DHS). Publications and work products need to be reviewed and approved by the DHS S&T ethics office and the NUSTL Director prior to public release.

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